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Laser ultrasonic imaging and damage detection for a rotating structure

Byeongjin Park¹, Hoon Sohn¹, Chul-Min Yeum² and Thanh C Truong¹

Abstract

This study presents a laser ultrasonic imaging and damage detection technique that creates images of ultrasonic waves propagating on a rotating structure and identifies damage. Laser ultrasonics is attractive for nondestructive testing mainly because of two reasons: (1) ultrasonic waves can be generated and/or measured in a noncontact manner and (2) even a small defect can be detected when laser ultrasonic scanning produces ultrasonic images with high spatial resolution. However, when it comes to a moving target, it becomes challenging to create reliable ultrasonic images. In this study, ultrasonic wave propagation images are obtained from a rotating blade using a pulse laser beam for ultrasonic generation, a galvanometer for laser scanning, and an embedded piezoelectric sensor for ultrasonic measurement. To properly estimate the laser excitation points during the scanning process rather than to precisely control the excitation points, a simple but rather effective localization technique is developed so that ultrasonic images can be constructed even from a moving target. Once the ultrasonic wave propagation images are created, damage on the target structure is visualized using a specially designed standing wave filter.

Keywords

Laser ultrasonics, rotating structure, localization, standing wave filter, damage visualization

Introduction

Rotating parts are critical components of various mechanical and structural systems but often prone to damage. For instance, the production of wind turbine blades is increasing with the growing market of the wind energy industry. Currently, the power generated by wind turbines is over 196 GW worldwide, and this number is increasing over 20% every year.¹ But it is reported that 13.4% of defects and 9.4% of downtime of the wind turbines are attributed to defects on wind turbine blades.² Various types of defects such as fatigue, fracture, buckling, and delamination occur on wind blades due to wind loading, lightening, and collision with foreign objects.^{3,4} Especially, delamination between laminates and adhesive layer disbonding are typical damage types for the wind blades made of composite materials.

There are a number of ongoing research efforts to monitor the structural safety and integrity of wind turbine blades while they are in operation. Vibration-based techniques attempt to evaluate the global integrity of the wind blades by relating the change of modal parameters such as natural frequencies, mode shapes, and

damping to the physical properties of the blades such as stiffness and mass.^{5,6} Fiber Bragg grating (FBG) strain sensors are embedded into the blades for the monitoring of wind blade bending,⁷ fatigue damage,⁸ and impact occurrence.⁹ Acoustic emission techniques detect crack initiation and localize it by measuring acoustic waves released by crack formation and growth.^{10–12} Using guided wave techniques, delamination as well as crack is detected by examining the interaction of propagating guided waves with defects.^{13–15} Here, guided waves are often generated and measured by piezoelectric transducers (lead zirconate titanates (PZTs)),¹³ air-coupled transducers,¹⁴ or wedge transducers.¹⁵

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For all of the aforementioned techniques, often a dense array of sensors is required in order to detect incipient and localized defects. However, the deployment of a large number of sensors and associated cabling become a daunting task, especially when it comes to a moving object like the wind blades. Furthermore, the interpretation of a plethora of data obtained from these sensors remains as a challenging task particularly in the presence of other operational and environmental variations.¹⁶ Finally, because these sensors are not defect-proof, the installation of increased number of sensors creates additional weak links and diminishes the overall system reliability, compromising the primary objective of structural health monitoring. In other words, the placement of too many sensors may end up deteriorating the overall system reliability rather than improving the system safety.

To address these issues, noncontact laser ultrasonic imaging techniques have been proposed, where ultrasonic waves are either generated or sensed by a scanning laser beam.^{17–22} The idea is to generate the images of propagating ultrasonic waves by exerting an input signal to a single discrete actuator and scanning corresponding responses using a sensing laser beam. Similar ultrasonic images can also be obtained by scanning an excitation laser beam and measuring resulting responses at a single sensing point. More recently, a fully noncontact laser scanning system is developed so that both the excitation and sensing laser beams can be scanned.²³ To obtain ultrasonic images using any of these scanning techniques, the position of the laser beam should be precisely controlled to intended target points. However, the application of the noncontact laser ultrasonic techniques to rotating objects has been limited because it is difficult to precisely aim a laser beam to a moving target point. For example, when a 63-m-long wind blade is rotating at 30 r/min, the speed of the blade tip is about 200 m/s and the tip deflection can be over 5 m.²⁴

In this article, a new laser ultrasonic imaging technique is proposed specifically for rotating blades. Ultrasonic waves are sequentially generated at multiple points using a scanning excitation laser beam, and corresponding responses are measured by a surface-mounted piezoelectric transducer at a single point. First, the excitation laser beam is approximately positioned near the target point using a galvanometer and then the precise position of the laser beam is estimated by an advanced localization technique. In addition, this study develops an advanced signal processing technique for automated visualization of a subsurface defect. The proposed laser ultrasonic technique is advantageous because (1) ultrasonic images with high spatial resolution can be obtained with scanning capability, making detection of incipient damage possible and subsequent image processing more intuitive and easier; (2) damage

diagnosis can be performed without using baseline data obtained from the pristine condition of a target structure, reducing false alarms due to changing environmental and operational conditions; and (3) only a small number of sensors are required for ultrasonic measurement, minimizing costs and time associated with sensor installation and maintenance.

This article is organized as follows: Section “Overview of noncontact laser ultrasonic scanning techniques” briefly overviews laser ultrasonic imaging techniques, and Section “The proposed laser ultrasonic imaging technique for a rotating blade” presents the proposed imaging technique for rotating structures with emphasis on the estimation of the laser excitation point. In Section “Laser ultrasonic imaging of a rotating blade and damage visualization,” the feasibility of the proposed laser ultrasonic technique is demonstrated by visualizing ultrasonic wave propagation on a rotating steel fan and detecting a defect. Finally, the conclusion and discussions are summarized in Section “Conclusion.”

Overview of noncontact laser ultrasonic scanning techniques

Laser ultrasonics has been around since 1960s and emerged as an effective tool for nondestructive testing (NDT) due to its noncontact nature and high-resolution scanning capability.^{17–22} Ultrasonic images can be constructed by exerting the ultrasonic excitation at a single point and measuring corresponding ultrasonic responses at multiple points using a scanning sensing laser beam. This scanning can be achieved by using a mechanical translator²⁵ or a scanning mirror.²⁶ The single point ultrasonic generation can be performed using a contact transducer such as a surface-mounted PZT or a noncontact laser beam as described later. Similar ultrasonic images can also be created by scanning the excitation laser beam instead of the sensing beam based on the linear reciprocity.²³ It can be easily shown that an ultrasonic signal excited at point A and measured at point B is identical to the one excited at point B and measured at point A. In practice, scanning the excitation laser beam is often more effective because laser sensing is highly dependent on the target surface condition and the incident angle of the sensing laser beam. Particularly, scanning of the excitation laser beam is attractive for curved surfaces or long-range scanning.

Ultrasonic waves can be generated using a high-power pulse laser such as Q-switched Nd:YAG and CO₂ lasers. When a target surface is exposed to a laser beam, a localized heating of the surface causes thermoelastic expansion of the material and generates ultrasonic

waves. Here, laser parameters such as the peak power, pulse duration, wavelength, and power density should be carefully adjusted not to cause surface damage called ablation.²⁷ Additionally, the surface condition, material property, and thickness should be considered for proper ultrasonic generation. For ultrasonic sensing, various laser interferometers such as homodyne, heterodyne, Mach-Zender, and Fabry-Perot techniques have been developed. A laser Doppler vibrometer, which is one of the heterodyne interferometers, is most widely accepted due to its high sensitivity and stability. This device measures the frequency shift of the laser beam when it reflects back from the target surface and relates the frequency shift to the out-of-plane velocity of the target surface based on the Doppler effect. Here, the angle of the incident laser beam with respect to the surface needs to be carefully adjusted to maximize the intensity of the laser beam returned along the direction of the incident laser beam, and often, a special surface treatment is necessary to minimize speckle noise.²⁸

Several scanning configurations have been applied to detection of defects in metal and composite structures. Examples of different configurations include excitation laser scanning and an embedded sensor at a fixed point,^{25,29} a surface-mounted ultrasonic actuator and sensing laser scanning,³⁰⁻³⁴ and a complete noncontact scanning system that can scan both the excitation and sensing laser beams.²³ Once ultrasonic wave propagation is visualized, the interaction of the propagating waves with the defect can be observed. However, the interaction can be blurred and smeared due to other waves and background noise, and the actual damage localization and quantification often demand human interpretation of the ultrasonic images. Several signal and image processing techniques have been proposed to accentuate the effect of damage and automate the damage localization and quantification processes.^{23,32,34}

Despite these remarkable advancements in laser ultrasonic imaging, little progress has been made for moving target imaging. When a fixed-point generation and sensing laser scanning is used, it is challenging to precisely aim the laser beam to a target point. More crucially, it is extremely difficult to measure reliable ultrasonic responses from a moving target. As an alternative, the excitation laser beam can be scanned, and the corresponding response can be measured using a fixed-point sensor. However, the precise control of the excitation laser beam still remains unsolved. Although several techniques are available for high-precision control of laser beams, these techniques tend to be rather expensive, complicated, and require marker installation on the target.³⁵⁻³⁷ In this study, we rather take a different approach. Instead of achieving high-precision control of the excitation laser beam, the laser beam is

roughly placed close to the target point and then the actual excitation point is more accurately estimated using the proposed localization technique. The details of the proposed localization technique are described in the next section.

The proposed laser ultrasonic imaging technique for a rotating blade

Overview of the proposed imaging technique

Figure 1 shows the overall schematic of the proposed laser ultrasonic imaging technique. The technique consists of five major steps: Step 1: collection of training signals from a stationary condition of the blade, Step 2: collection of an inspection signal from a rotating condition, Step 3: estimation of the laser excitation point by computing correlations between the inspection signal and each of the training signals, Step 4: construction of ultrasonic images by repeating Steps 2 and 3 for all the entire inspection region during the blade rotation, and Step 5: visualization of the damage by applying an image processing technique to the ultrasonic images constructed in Step 4.

Step 1 is a process to collect training data from a stationary condition of the blade. When an excitation laser beam is exerted to a specific point on the blade with known coordinates, the corresponding ultrasonic response signal, named a training signal, is measured from an ultrasonic sensor mounted on the blade. This process is repeated over the entire training grids by scanning the excitation laser beam and measuring the training signals from the ultrasonic sensor. The accuracy of the laser excitation point estimation heavily depends on the grid spacing in training. The more densely the training data are collected, the more precisely the excitation laser beam location is estimated. Therefore, the grid spacing needs to be close to the spatial resolution required for ultrasonic imaging. For example, to obtain the ultrasonic images with spatial resolution of 2 mm, the training grid spacing is set to be 2 mm in the later experiment.

Once the training is completed and the blade is in operation, inspection starts by firing the excitation laser beam and measuring the corresponding ultrasonic response, named an inspection signal (Step 2). Here, the position of the excitation laser beam is controlled so that it can be aimed as closely as possible to the intended target position, but there will be always some discrepancy between the target and actual excitation positions because of blade rotation and ambient vibration. Therefore, the precise excitation point needs to be estimated in the following step with accuracy comparable to the spatial resolution required for ultrasonic imaging.

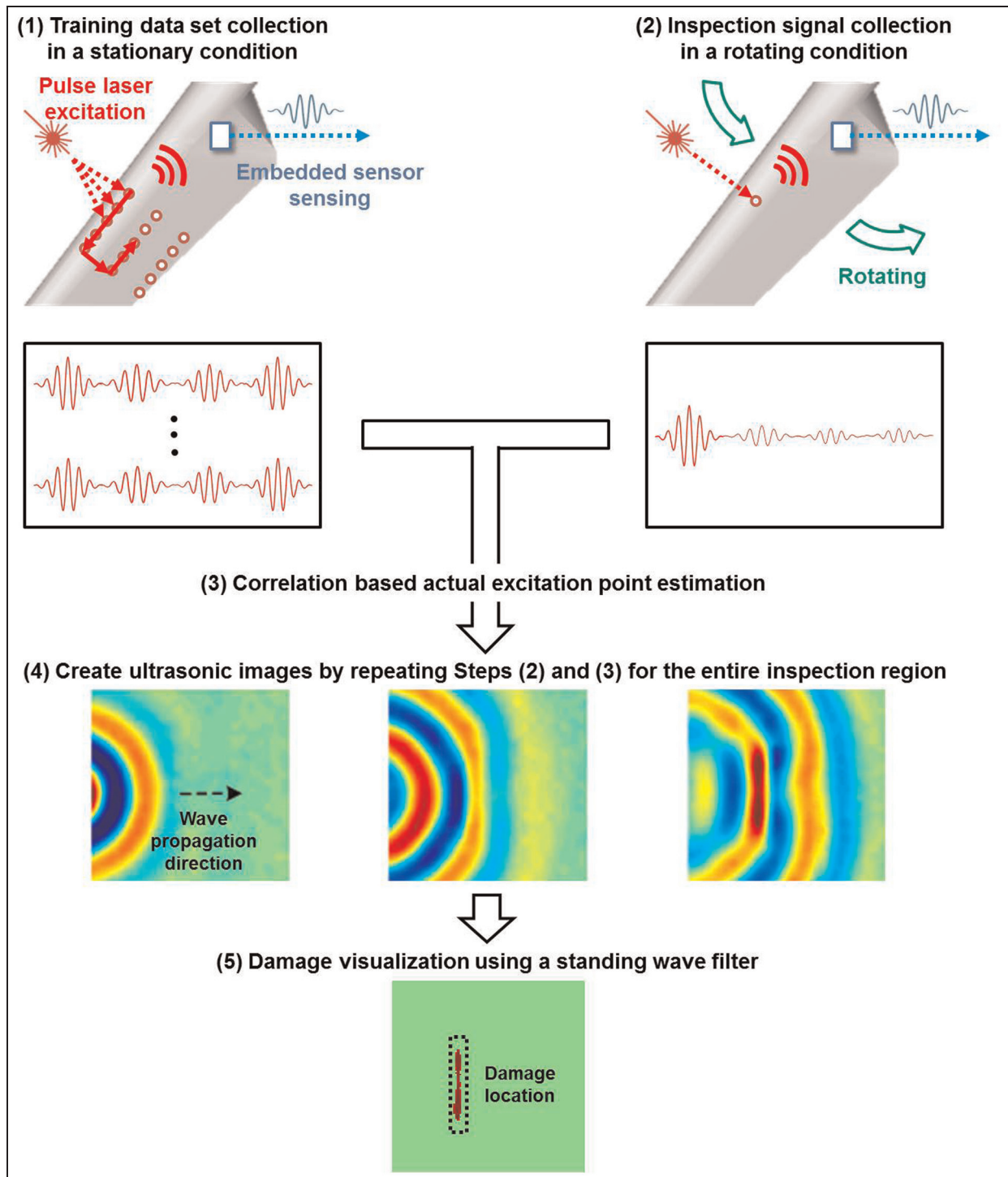


Figure 1. Overall schematic representation of the proposed laser ultrasonic imaging technique for a rotating blade: (1) training signals are obtained from a stationary condition of the blade by scanning the excitation laser beam over the entire training region and measuring the corresponding responses using an embedded sensor; (2) for actual inspection, an excitation laser beam is exerted to the blade and the accompanying ultrasonic response is measured when the blade is rotating; (3) the exact excitation point is estimated by computing correlations between the inspection signal obtained from the rotating blade with those in the training data set. The training signal that has the maximum correlation with the inspection signal reveals the actual position of the inspection laser beam excitation; (4) Steps 2 and 3 are repeated by scanning the excitation laser beam over the entire inspection region, and ultrasonic images are obtained using the estimated excitation points and measured responses; and (5) by applying a standing wave filter, the damage location can be visualized from the ultrasonic images.

In Step 3, the actual excitation position is estimated by computing correlations of the inspection signal with each of the ultrasonic signals stored in the training data set. Let $f(t)$ and $g(t)$ represent the inspection and training signals, respectively. Then the normalized correlation between two signals is represented as follows

$$(f \star g)(\tau) = \frac{\int_{-\infty}^{\infty} f(t)g(\tau+t)dt}{\sqrt{\int_{-\infty}^{\infty} f^2(t)dt \cdot \int_{-\infty}^{\infty} g^2(t)dt}} \quad (1)$$

where $\star$ denotes the correlation operator. When the coordinates of the laser excitation point associated with any of the training signals become identical to those of the inspection signal, the maximum value of $(f \star g)(\tau)$, denoted as a localization index (LI), becomes 1. Therefore, the coordinates of the training signal, which has the maximum LI value with the inspection signal, are identified as the most likely excitation point. This concept is based on time reversal, and applied to impact localization,^{38,39} localization of an acoustic emission source,⁴⁰ and estimation of a laser excitation point.⁴¹

Once the laser excitation point is estimated, the inspection signal is registered as the response corresponding to the estimated coordinates. Then, Steps 2 and 3 are repeated by scanning the excitation laser beam over the entire inspection region, and the ultrasonic images covering all the inspection surfaces are obtained as a function on time (Step 4). When there is more than one inspection signal assigned to the same location, the responses are averaged.

In Step 5, the obtained ultrasonic images are treated with advanced image processing for damage visualization. When incident waves encounter a defect, multiple reflections are generated, and the superposition of the incident and reflection waves produces standing waves. A standing wave filter previously developed by the authors' group is used to isolate only these standing wave components trapped inside the defect.²³ First, the propagating waves in the obtained ultrasonic images (W_T) are decomposed into forward (W_F) and backward (W_B) propagating waves using a frequency-wavenumber domain analysis. Then, standing wave energy (SWE) is obtained by subtracting the energies of the decomposed forwarding and backward waves from the total wave energy, $SWE = W_T^2 - W_F^2 - W_B^2$. Finally, the cumulative SWE (CSWE) image is constructed as $CSWE(x, y, t) = \int_0^t SWE d\tau$ for all spatial points. As the standing waves are produced near the damage location, the accumulation of SWE is visualized using CSWE. More details can be found in the previous work of the author's group.²³

Consideration of damage and temperature effects

In Step 3, the laser excitation point is estimated by computing correlations of the inspection signal with each of the ultrasonic signals stored in the training data set. In practice, the velocity, amplitude, and scattering of ultrasonic waves are heavily influenced by the operational and environmental conditions such as temperature.⁴² Therefore, if the inspection signal was obtained after the appearance of damage or from a temperature condition significantly different from the conditions where the training data were collected, it would be no longer identical to the training signal corresponding to the same excitation point.

The effect of damage on localization is addressed by applying a band-pass filter to both the inspection and training signals so that only certain low-frequency components of the filtered signals are used for correlation calculation. Here, the use of only low-frequency contents can be justified because often high-frequency ultrasonic responses are more sensitive to local defects than low-frequency components.⁴³

Another source of the localization error is temperature variation. When two ultrasonic signals are obtained from a same position but from different temperature conditions, one of the signals will be delayed with respect to the other signal, and this time delay tends to increase along the time axis. To alleviate the time delay effect due to temperature variation, $f(t)$ and $g(t)$ are first divided into multiple segments, $f_i(t)$ and $g_i(t)$, using a moving time window. $f_i(t)$ and $g_i(t)$ represent the i th segment of the inspection and training signals, respectively. Then, the LI between $f_i(t)$ and $g_i(t)$ are computed similar to equation (1). The computation of the normalized correlation between the two segments is repeated by shifting the window with a certain overlap until it reaches the end of the signals. Finally, the compensated $\overline{LI}$, which is the root mean square (RMS) value of the maximum correlation values, is obtained as follows

$$\overline{LI} = \sqrt{\frac{1}{N} \left(\sum_{i=1}^N \text{MAX}\{(f_i \star g_i)(\tau)\}^2 \right)} \quad (2)$$

where N represents the total number of segments in each signal. This process is summarized in Figure 2. Using $\overline{LI}$, the effect of temperature variation on the estimation of the laser excitation point can be alleviated although it cannot be fully eliminated. Then the point with the highest $\overline{LI}$ is identified as the most likely laser excitation location.

Note that the aforementioned parameters, including the filter range, the size of the moving window, and the overlap percentage between two subsequent

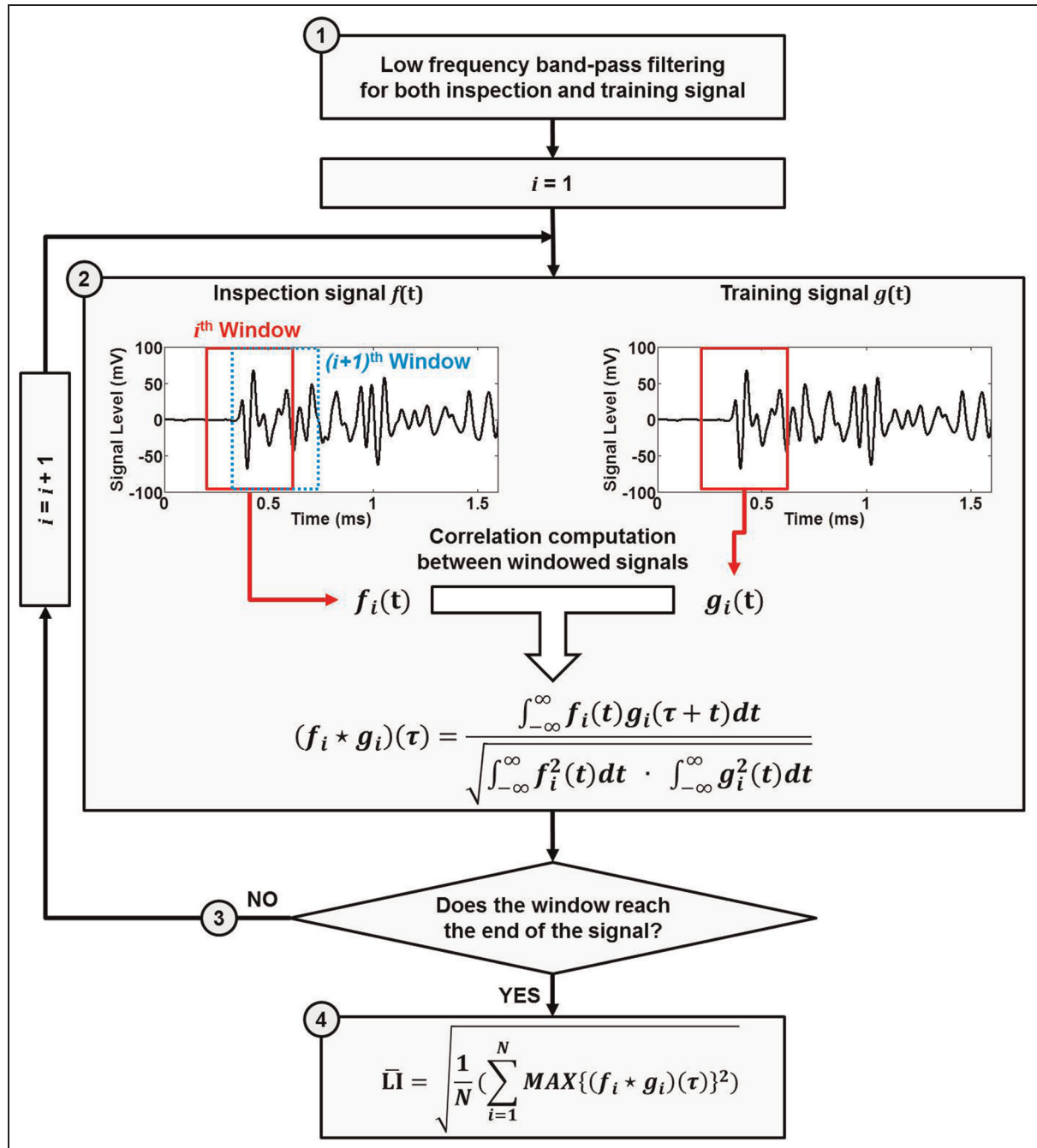


Figure 2. Computation of the compensated localization index ($\bar{LI}$): (1) to minimize the effect of damage on localization, the inspection and training signals are band-pass filtered so that only a certain low-frequency content of the signals are used for localization; (2) the correlation between the i th windowed segments obtained from the filtered inspection and training signals is computed; (3) this process is repeated until the moving window reaches the end of the signal; and (4) finally, $\bar{LI}$ is obtained as the RMS value of the computed correlation values.

moving windows, are problem specific. The performance of the proposed localization technique depends on the optimal selection of these parameters, and they could be adjusted through careful considerations of the geometry and materials of the target structure.

Laser ultrasonic imaging of a rotating blade and damage visualization

Experimental setup

The effectiveness of the proposed laser ultrasonic imaging technique is examined using an industrial steel fan

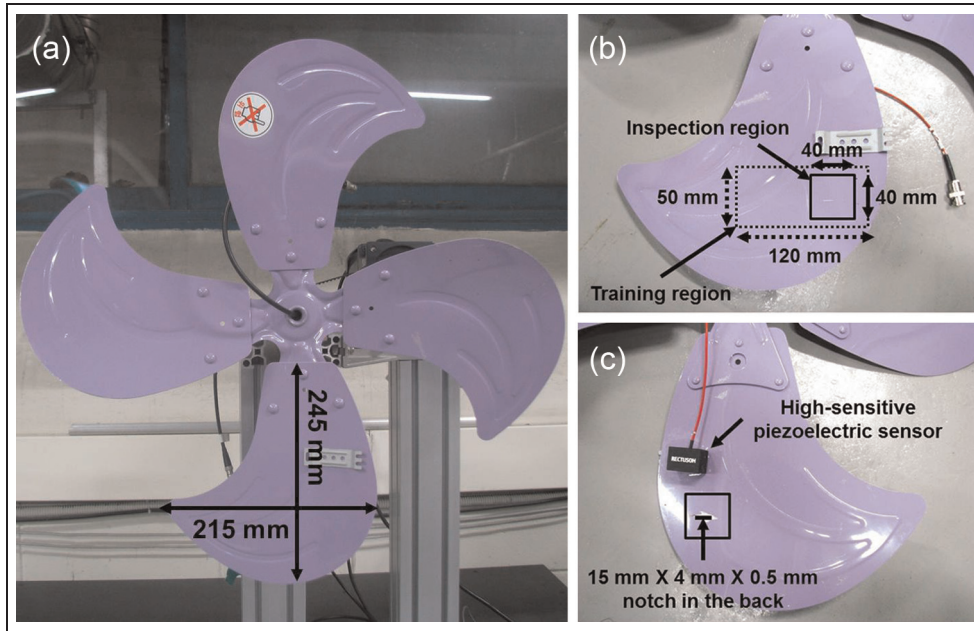


Figure 3. A rotating steel fan with a high-sensitive piezoelectric sensor installed on the back side of one of the blades. The dotted and solid boxes in (b) indicate training and inspection regions, respectively: (a) a steel fan with complex geometry and pressed patterns, (b) front side of the blade, and (c) rear side of the blade with a sensor.

as shown in Figure 3. The fan has four steel blades, and ultrasonic images are taken from one of the blades. Each blade is 24.5 cm long, 21.5 cm wide, and 1 mm thick. Each blade has a complex geometry, and the blade surface is pressed with some patterns. In Figure 3(b), the training and inspection regions are marked with the dotted and solid boxes, respectively. There are 1386 grid points inside the 50 mm by 120 mm training region with grid spacing of 2 mm, and 400 grid points inside the 40 mm by 40 mm inspection region with the same grid spacing. A notch with 15 mm length, 4 mm width, and 0.5 mm thickness is introduced on the rear side of one of the blades (Figure 3(c)).

To realize the proposed laser ultrasonic imaging technique, a measurement system is set up as shown in Figure 4. For the noncontact ultrasonic wave generation, a Nd:YAG pulse laser with 6 ns pulse duration (Quantel Ultra laser) is used. The energy of the pulse laser is restricted to 5 mJ to avoid ablation damage on the specimen surface. A galvanometer (Scanlab Scancube10), composed of two rotating mirrors, is synchronized with the pulse laser to allow two-dimensional (2D) scanning of the laser beam. To synchronize the rotation of the target structure with the firing of the excitation laser beam, an Autronics E40H6 encoder is installed on the motor. The generated ultrasonic waves are measured with a Fujicera M204A high-sensitive piezoelectric sensor, which is installed on the back surface of the target blade. The measured ultrasonic waves are transmitted to a data acquisition system (DAQ)

through a slip ring installed in a motor shaft. Finally, the DAQ (NI PXI-5122) collects and stores the measured signals for the ultrasonic imaging process. The rotating speed of the specimen is approximately 20 r/min.

Data collection and processing

During data collection process, each ultrasonic signal is collected over 3.2 ms with a sampling frequency of 5.12 MHz, resulting in 16,384 time points per each signal. For the training, 20 ultrasonic signals corresponding to each training grid point are averaged to improve the signal-to-noise ratio. For inspection, the excitation laser beam is repeatedly shot at each grid point 30 times, and the corresponding responses are stored. The actual excitation position of each shot, which varies every time due to the rotation and vibration of the blade, is estimated using the previously described localization technique. Although laser excitation is repeated 12,000 times (400 grid points $\times$ 30 shots for each point) aiming for the inspection region, only 2/3 of them actually hit the intended region and the rest fall outside the inspection region. Overall, each grid point is hit by the laser beam 20 times, and the responses at each grid point are averaged. Once ultrasonic responses are obtained from all inspection grid points, the ultrasonic images are constructed for the inspection region. Note that because the laser beam may be shot outside the inspection region as the fan rotates with a high speed,

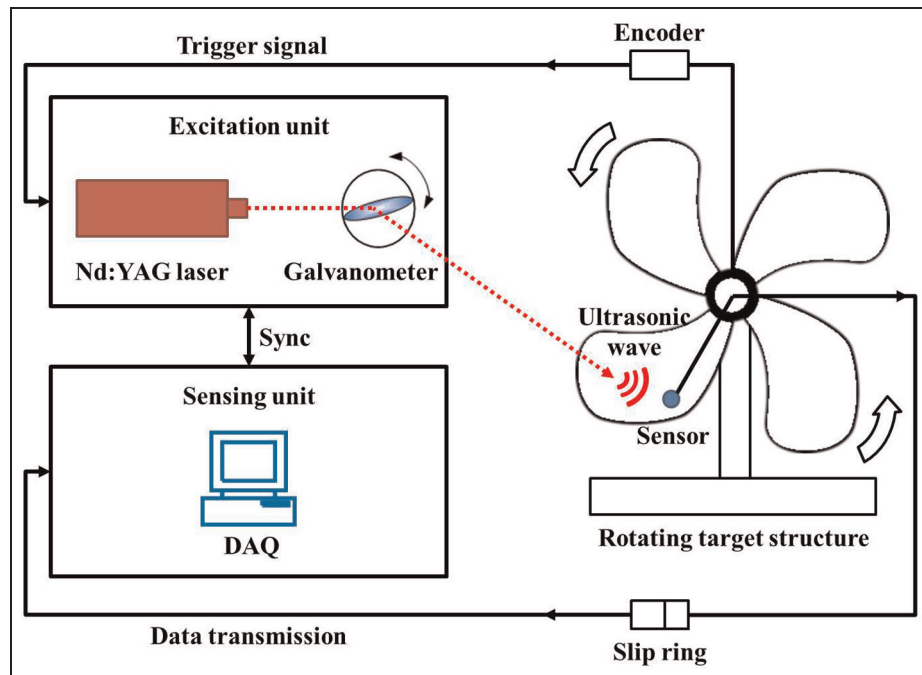


Figure 4. Schematic diagram of the overall experimental setup. A Nd:YAG laser beam is fired when the encoder detects a single rotation of the specimen. The corresponding ultrasonic waves are measured by a surface-mounted sensor and transmitted to a data acquisition system (DAQ) through a slip ring. Nd:YAG laser scanning is achieved using a galvanometer, and the excitation unit and the sensing unit are synchronized to each other.

Table 1. Validation of the excitation point estimation technique.

	Mean estimation error (mm)
Case 1 (25°C without damage)	0.0
Case 2 (45°C without damage)	0.6
Case 3 (25°C with damage)	0.3
Case 4 (45°C with damage)	1.2

The excitation points are successfully estimated for all cases, and the average estimation errors for all cases are below the grid spacing of 2 mm.

the training region is set larger than the inspection region.

For the estimation of the excitation laser point, both the training and inspection signals are processed with a band-pass filter with lower and upper cutoff frequencies of 1 and 20 kHz, respectively. Then, a 600- μ s moving window with 20% overlap is applied for the calculation of $\overline{LI}$ in equation (2). For ultrasonic imaging, each raw inspection signal is band-pass filtered, and the response only between 130 and 200 kHz is visualized. Note that the lower frequency response between 1 and 20 kHz is used for laser excitation point localization and the higher frequency response in the range of 130 and 200 kHz for ultrasonic imaging and damage visualization.

Experimental results

Excitation point estimation. The performance of the proposed excitation point estimation technique is examined for four different scenarios as shown in Table 1 considering temperature and damage effects: Case 1: inspection signals are obtained at room temperature (25°C) without any damage on the blade, Case 2: collected at 45°C without damage, Case 3: collected at 25°C with damage, and Case 4: collected at 45°C with damage. The temperature of the blade is controlled with a ceramic heater placed on the back of the blade, and measured by a noncontact infrared (IR) thermometer. The training data are collected from the training region at 25°C. For each test case, the excitation laser beam is exerted to 400 grid points within the inspection region when the blade is in a stationary condition, and the position of each laser excitation is estimated using the proposed estimation technique. Then, the estimation error between the actual and estimated excitation points is computed for each grid point, and the mean estimation error is computed by averaging the estimation error over all grid points. Note that the test signals in this particular experiment are obtained when the steel fan is in a stationary condition, otherwise the exact laser excitation points are unknown and the estimation error cannot be computed.

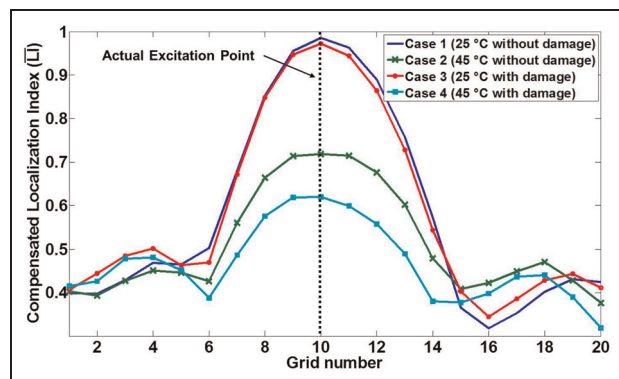


Figure 5. Distribution of $\bar{LI}$ for each four case. The actual excitation point (grid number 10, which is indicated by a dashed line) has highest $\bar{LI}$ and correctly estimated.

Figure 5 shows a spatial distribution of $\bar{LI}$ for four different cases. The $\bar{LI}$ value dramatically decreases as it moves further away from the actual excitation point (grid number 10).

The results of the excitation point estimation are summarized in Table 1. Overall, the actual excitation points are well identified under temperature variation and in the presence of damage. When the test data are obtained from the same condition as the training data, the excitation points are exactly identified (Case 1). With temperature increase (Case 2) and damage appearance (Case 3), the estimation error increases. However, even with both temperature change and damage (Case 4), the average estimation error of 1.2 mm is below the required spatial resolution of ultrasonic imaging, which is the grid spacing of 2 mm.

Ultrasonic imaging of a rotating blade. Figure 6 shows the images of propagating ultrasonic waves obtained from the inspection region of the blade shown in Figure 3(b) when there is no defect on it. The red and blue colors in the images represent positive and negative out-of-plane velocities, respectively, in a linear scale (color online). Ultrasonic snapshots obtained from a stationary condition of the blade at 25, 35, and 45 μ s are shown in Figure 6(a) (a video clip is available in the online version of the article). Then, the ultrasonic images are obtained from the same inspection region again, but this time, when the blade is rotating. Snapshots corresponding to the same time points as before are shown in Figure 6(b). Because of the rotation of the blade as well as additional vibration, the excitation laser beam could not be precisely positioned to the target points only by controlling the scanning mirror. This misalignment of the laser excitation point with the target point results in the distortion of ultrasonic wave propagation images in comparison with the ones in Figure 6(a). On

the contrary, it is shown in Figure 6(c) that the ultrasonic images are successfully reconstructed by employing the proposed localization technique.

To quantify the performance of the ultrasonic image reconstruction using the proposed technique, 2D cross-correlation between two ultrasonic images are calculated, and the RMS value of the correlation is computed for the entire time duration of the video clip. The RMS correlation between the ultrasonic images obtained from the stationary and rotating conditions improves from 66.6% in Figure 6(b) to 98.4% in Figure 6(c) by employing the proposed localization technique.

In Figure 7, the same experiments as the previous ones shown in Figure 6 are repeated after introducing a defect to the blade. The overall trend is consistent with the previous experiments. The ultrasonic propagation video in Figure 7(c) quite well matches with the one in Figure 7(a) with 96.2% correlation, while Figure 7(b) has only 67.4% correlation. However, the correlation for the damage case, that is, the correlation between Figure 7(a) and (c), decreases compared to the counterpart correlation of the intact case, the correlation between Figure 6(a) and (c), especially near the damage location. This decay of the correlation is attributed to the fact that the estimation error between the actual and estimated excitation points increases near the damage and location. In fact, the increased estimation error near the damage can affect the subsequent damage visualization in our favor, increasing signal distortion near the damage. Furthermore, scattering and multiple reflections are observed near the damage as shown in Figure 7(a) and (c).

Damage visualization. Here, additional image processing is performed on the raw ultrasonic images obtained from the rotating blade with damage. For easier comparison, the snapshot at 35 μ s shown in Figure 7(c) is reproduced in Figure 8(a). When the propagation of ultrasonic waves is displayed as a function of time, the interaction of propagating waves with the defect is better observed although it is not clearly shown in the snapshot taken in Figure 8(a).

Then, the propagation of the cumulative ultrasonic wave energy is visualized by computing the RMS value of the ultrasonic wave signal at each spatial point as a function of time.³² Because of multiple reflections and scattering near the defect, increased energy concentration is expected to occur near the damage. As expected, the surge of the cumulative ultrasonic energy is observed near the damage in Figure 8(b). However, due to the complex patterns pressed on the blade and multiple reflections from the boundaries, additional energy concentrations are also observed outside the

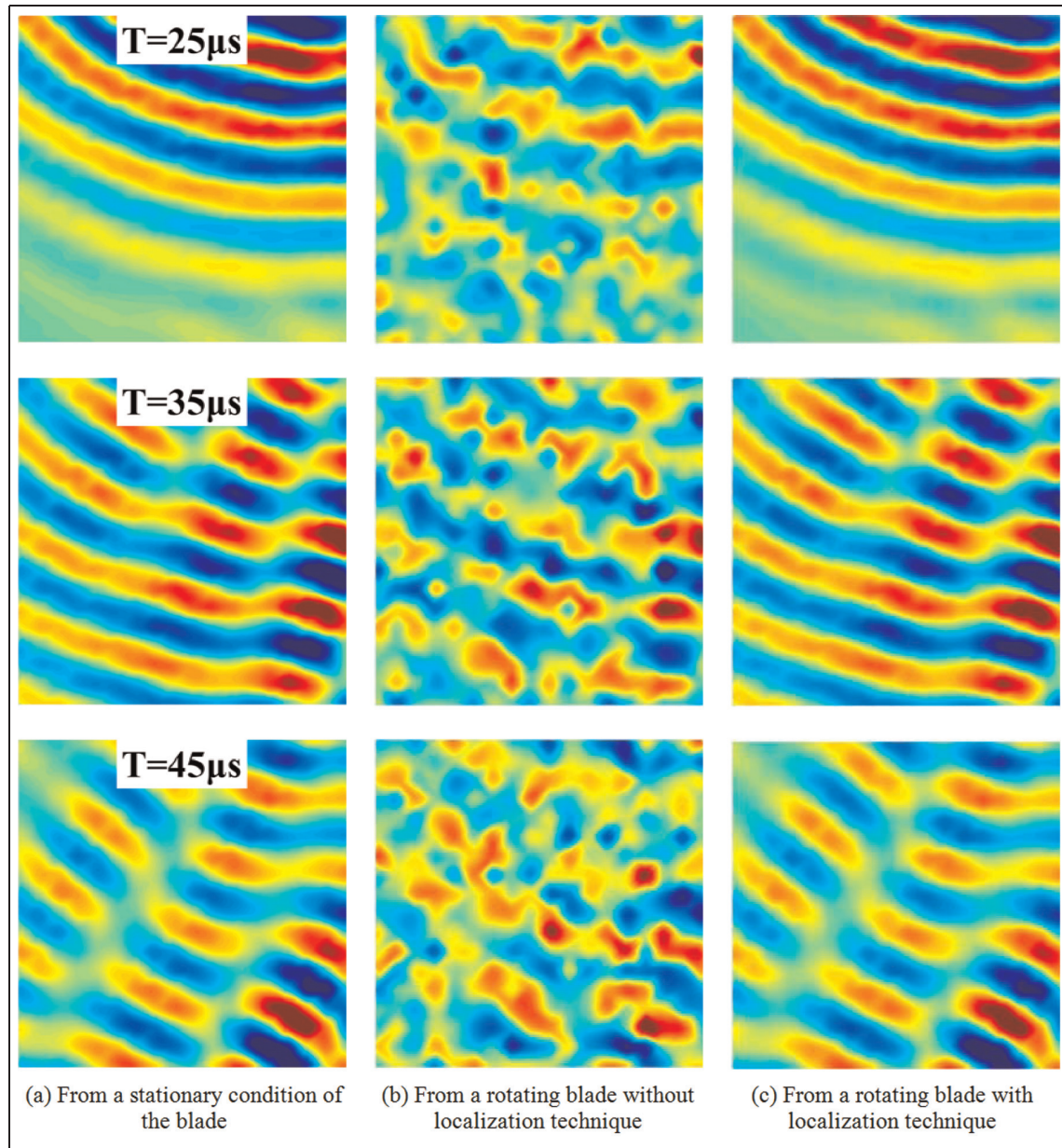


Figure 6. Ultrasonic wave propagation images obtained from an intact blade at 25, 35, and 45 μs (video clip available in the online version). The image corresponds to the inspection region shown in Figure 3(b). Red and blue colors correspond to positive and negative out-of-plane velocities, respectively (color online). The correlations of (b) and (c) with the reference video clip of (a) are 66.9% and 98.4%, respectively: (a) from a stationary condition of the blade, (b) from a rotating blade without localization technique, and (c) from a rotating blade with localization technique.

damage area. In Figure 8(b), the red color indicates the high-energy concentration areas.

To better isolate the effect of the damage, a standing wave filter previously developed by the authors' group is applied to the raw ultrasonic image.²³ It has been shown that ultrasonic waves are trapped inside a defect, propagating in opposite directions and resulting in standing waves. The standing wave filter extracts these standing wave components from the measured raw

ultrasonic signals, and the resulting CSWE image is shown in Figure 8(c). Mostly, the damage location is highlighted in the image, enabling intuitive damage visualization.

Conclusion

In this study, a new laser ultrasonic imaging technique is developed specifically for a rotating structure.

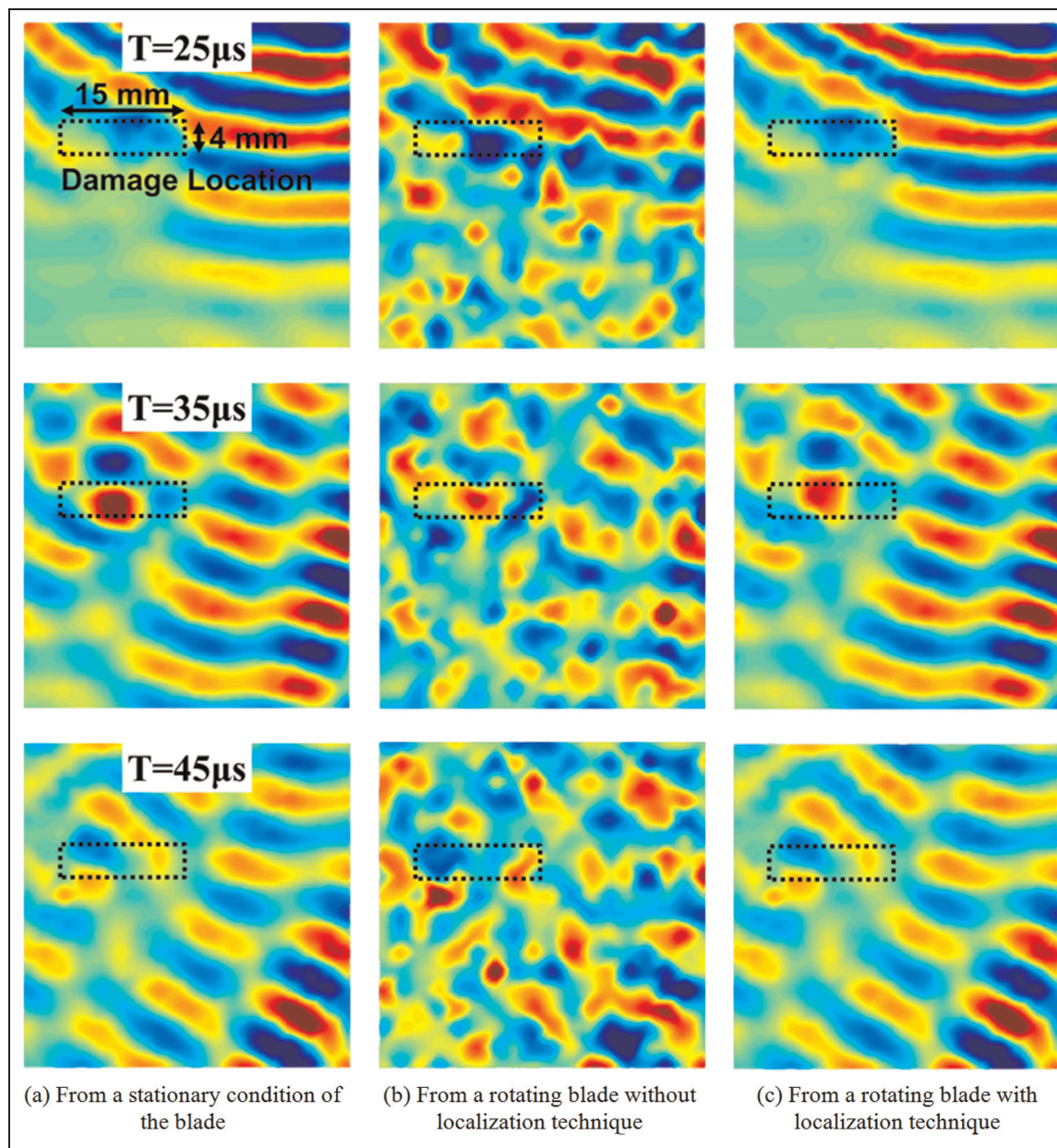


Figure 7. Ultrasonic wave propagation images obtained from the damaged blade at 25, 35, and 45 μs (video clip available in the online version). The image corresponds to the inspection region shown in Figure 3(b). Red and blue colors correspond to positive and negative out-of-plane velocities, respectively (color online). The correlations of (b) and (c) with the reference video clip of (a) are 67.4% and 96.2%, respectively: (a) from a stationary condition of the blade, (b) from a rotating blade without localization technique, and (c) from a rotating blade with localization technique.

Ultrasonic waves are generated using a scanning Nd:YAG pulse laser, and the associated ultrasonic responses are measured using a piezoelectric transducer mounted on the target structure. Because it is difficult to exactly control the excitation position of the scanning laser beam when the laser beam hits a rotating target structure and creates ultrasonic waves, the proposed technique first roughly aims the excitation laser beam

to a target point and then more precisely estimates the actual excitation point. The essence of the proposed technique is to pinpoint this laser excitation position even when the target structure is rotating with temperature variation and the possibility of damage presence. The effectiveness of the proposed imaging technique is examined by visualizing ultrasonic wave propagation on a rotating steel blade. It has been demonstrated that

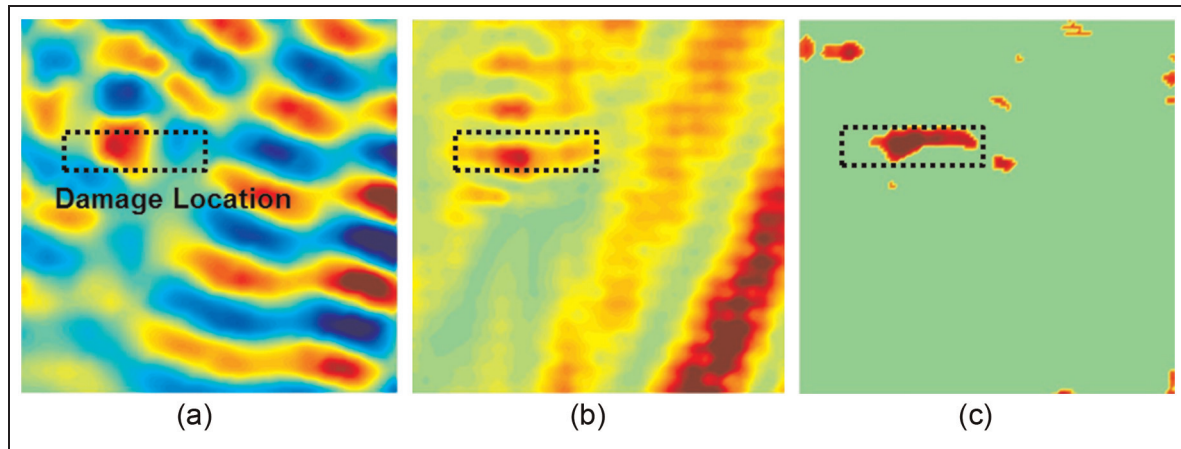


Figure 8. Damage visualization from the previous ultrasonic images. With the aid of the standing wave filter, the damage effect is highlighted in the image. A linear scale is used to represent each image: (a) raw wave propagation image, (b) cumulative energy propagation image, and (c) cumulative standing wave energy propagation image after standing wave filtering.

the ultrasonic images can be successfully constructed in spite of the fast rotating speed and the complex geometry of the blade. The damage invisible from the scanned surface has been also successfully identified, and its visibility has been enhanced using the standing wave filter previously developed by the authors' group.

Further studies are warranted before the proposed technique can be applied for more realistic structures. First, the proposed technique, as it is, still requires the placement of a wired sensor for ultrasonic measurement. Therefore, the inspection region is limited, and a large number of sensors need to be installed to inspect a large area. An alternative is to use a noncontact laser interferometer for ultrasonic measurement, but its application to a moving object is limited at this point. Second, the proposed localization technique requires a prior collection of the training data from a stationary condition of the target structure. As the target structure becomes large, scanning takes a longer time, the amount of the training data increases, and the subsequent computation becomes more intensive. Follow-up studies are underway to address these issues.

Declaration of conflicting interests

The authors declare that there is no conflict of interest.

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